

EXPERIMENT NO : 10

Experiment to understand array function in Data science.

PROGRAM:-

```
import numpy as np
```

```
array=np.random.randint(1,100,9)
```

```
array
```

OUTPUT:-

```
array([44, 2, 19, 8, 60, 12, 42, 51, 86])
```

```
np.sqrt(array)
```

OUTPUT:-

```
array([6.63324958, 1.41421356, 4.35889894, 2.82842712, 7.74596669,  
3.46410162, 6.4807407 , 7.14142843, 9.2736185 ])
```

```
array.ndim
```

OUTPUT:-

```
1
```

```
new_array=array.reshape(3,3)
```

```
new_array
```

OUTPUT:-

```
array([[44, 2, 19],  
       [ 8, 60, 12],  
       [42, 51, 86]])
```

```
new_array.ndim
```

```
2
```

```
new_array.ravel()
```

OUTPUT:-

```
array([44, 2, 19, 8, 60, 12, 42, 51, 86])
```

```
newm=new_array.reshape(3,3)
```

```
newm
```

OUTPUT:-

```
array([[44, 2, 19],  
       [ 8, 60, 12],  
       [42, 51, 86]])
```

```
newm[2,1:3]
```

OUTPUT:-

```
array([51, 86])
```

```
newm[1:2,1:3]
```

OUTPUT:-

```
array([[60, 12]])
```

```
new_array[0:3,0:0]
```

OUTPUT:-

```
array([], shape=(3, 0), dtype=int64)
```

```
new_array[0:2,0:1]
```

OUTPUT:-

```
array([[44],  
       [ 8]])
```

```
new_array[0:3,0:1]
```

OUTPUT:-

```
array([[44],  
       [ 8],  
       [42]])
```

```
new_array[1:3]
```

OUTPUT:-

```
array([[ 8, 60, 12],  
       [42, 51, 86]])
```